



GRUMMAN X-29

BY JAMES BRAY



INITIAL CONCEPT

At the height of the Cold War, the X-29 was conceptualised by the U.S. Air Force and funded by **DARPA** (Defense Advanced Research Projects Agency), the **USAF** and **NASA**.

In an attempt to build the ‘ultimate fighter jet,’ DARPA created an experimental design with **forward-swept wings**.

Although this had been done successfully before in 1944 by **Hans Wocke** DARPA wanted to weaponise this concept for their military applications



DESIGN



3 Surface Aircraft

Canards

Small forewings placed in front of the main wings for stabilisation

Forward Swept Wings

Generate high moments for agility/manoeuvres and allow for a greater aspect ratio

Aft Fuselage Strakes

Reduce drag and improve take off and landing stability

Aircraft Specs

Length: 53 ft

Wingspan: 27 ft

Sweep Angle: 33°

Height: 14 ft

Wing area: 17.544 m²

Aspect ratio: 3.9

Empty weight: 6,260 kg

Max takeoff weight: 8,074 kg

Engine: General Electric

F404 turbofan engine

Engine Thrust: 71 kN

Total Cost: \$87 million

DESIGN



Components

Nose Landing Gear & Front Fuselage

Taken from the F-5A
Freedom Fighter

Control surface actuators
Taken from the F-16

Wing Material

Consists of a graphite epoxy which is able to cope with high pitching moments on the wings

Main Landing Gear
Taken from the F-16

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DIFFICULTIES

The forward-swept wings lead to an increased lift on the leading edge and **increased pitching moments** on the wings

Due to the advancements in carbon fibre composites, these wings could now be made from **graphite epoxy**.

The X-29's centre of gravity turned out to be far behind the aerodynamic centre, resulting in the aircraft being **inherently unstable**.



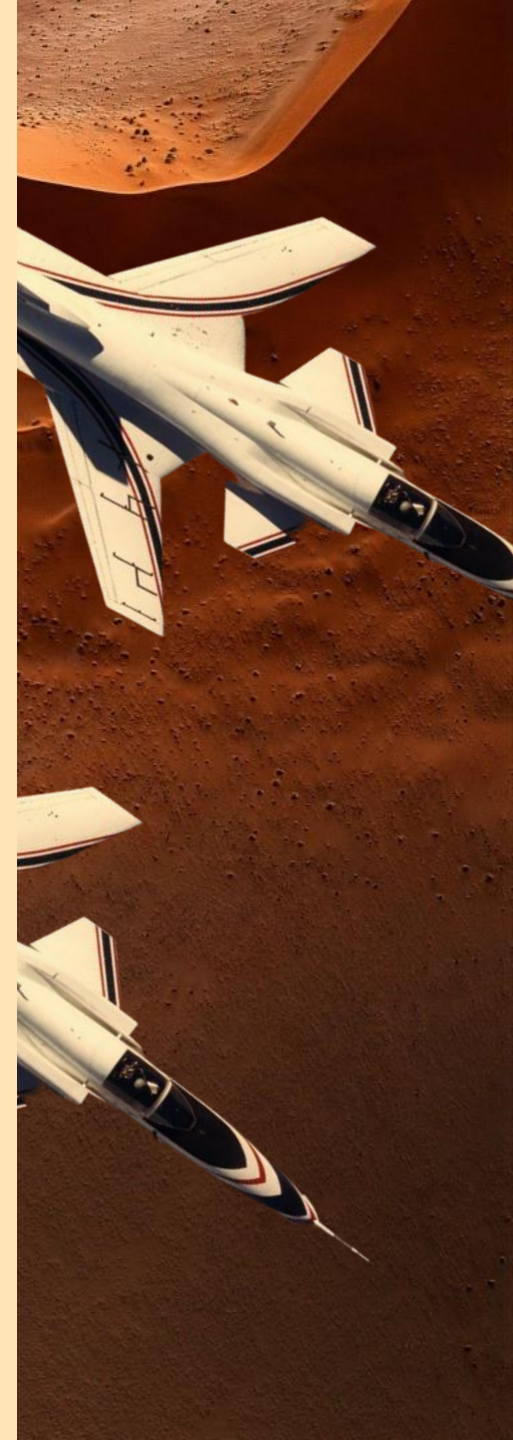
STABILISATION

*“An F-18 fighter jet has an instability factor of only 5%.
The X-29, on the other hand, was 35% unstable”*



To stabilise the aircraft, DARPA installed **3 flight control systems** which corrected the flight path **40 times/second**.

If all three were to fail simultaneously, the X-29 would tear up before the pilot had a chance to eject

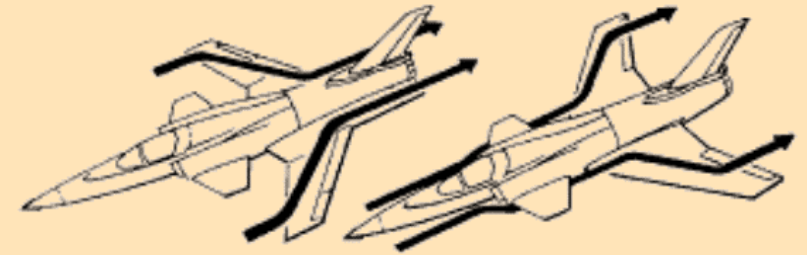


FLIGHT

The aircraft first flew on December 14th 1984, and two X-29s were flight tested through 1991.

Most flights were research flights experimenting with stall limits and max speeds

Forward-swept wings mean stalls start closer to the fuselage, leaving the ailerons functional for longer and giving pilots time to regain control.



The X-29 was the first supersonic FSW aircraft



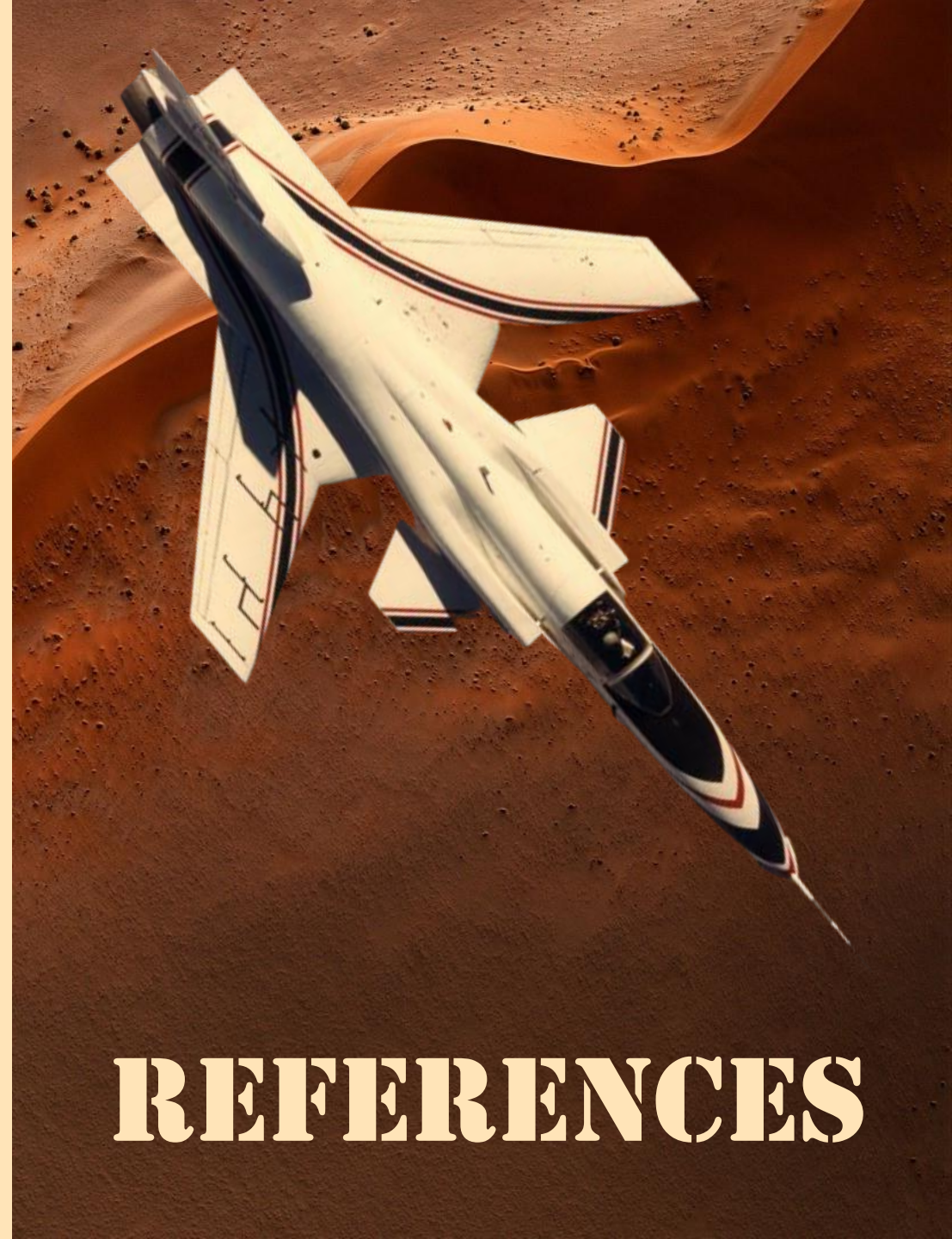
THANK YOU

[Wikipedia](#)

[National Museum of the US Air Force](#)

[The X-29 — A Unique and Innovative Aerodynamic
Concept](#)

[CNN Style](#)



REFERENCES